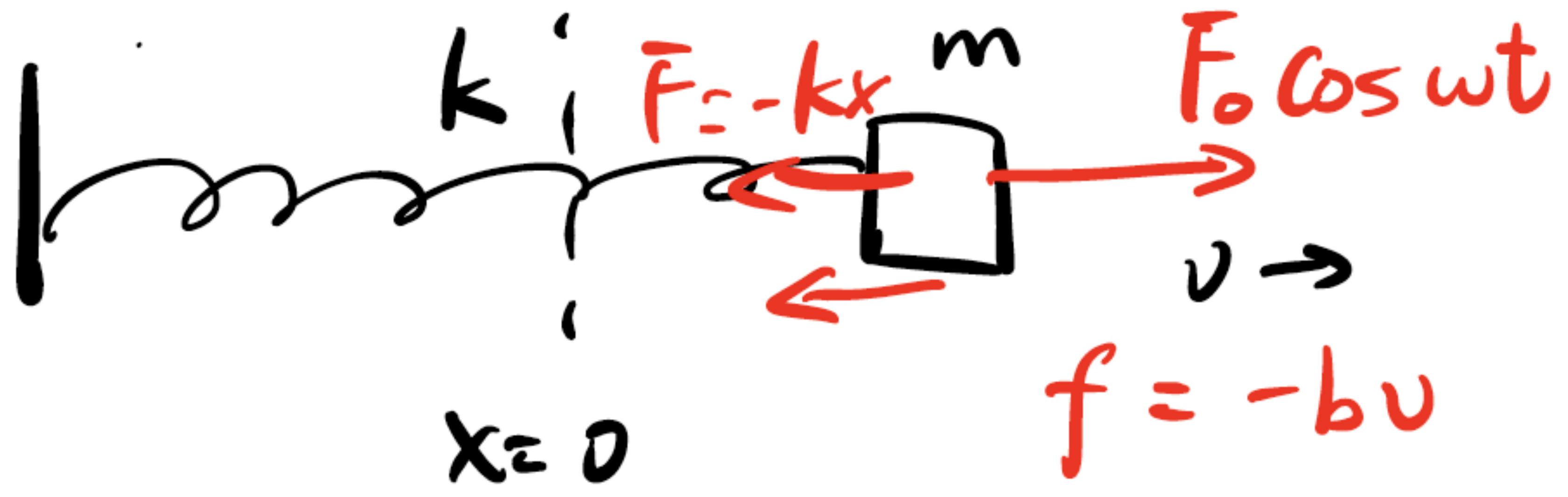
香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

Lecture 26: Wave

Jingqiang Ye
School of Science and Engineering (SSE)
April 24th, 2025

Recap: Oscillation



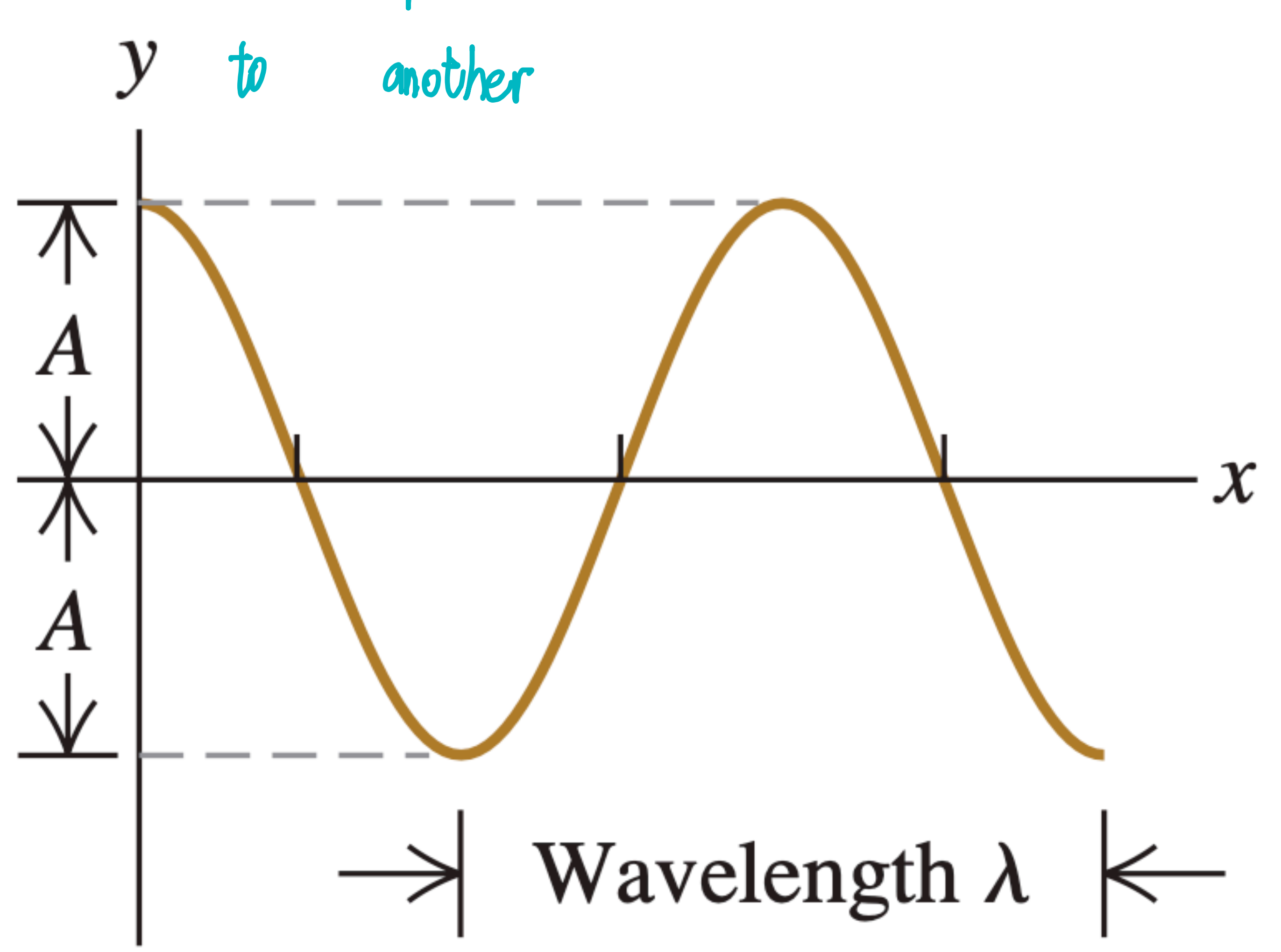
$\gamma = \frac{b}{m}$, $\omega_0 = \sqrt{\frac{k}{m}}$

- Forced Oscillation w/ Damping
  - Equation of motion (EOM):
  - $ma = -kx - bv + F_0 \cos \omega t$
  - $\frac{d^2x}{dt^2} + \omega_0^2 x + \gamma \frac{dx}{dt} - \frac{F_0}{m} \cos \omega t = 0$

Recap: Wave

- Transverse & longitudinal wave
- Wave function $y(x, t) = A\cos(kx - \omega t)$, where
  - A, amplitude
  - $k = \dfrac{2\pi}{\lambda}$, (angular) wave number
  - $\omega = \dfrac{2\pi}{T}$, angular frequency
- Wave equation $\dfrac{\partial^2 y}{\partial t^2} = v^2 \dfrac{\partial^2 y}{\partial x^2}$
- Wave speed $v = \dfrac{\lambda}{T} = \dfrac{\omega}{k} = \sqrt{\dfrac{F_T}{\mu}}$, where
  - μ: linear density of string
  - F_T: tension in the string

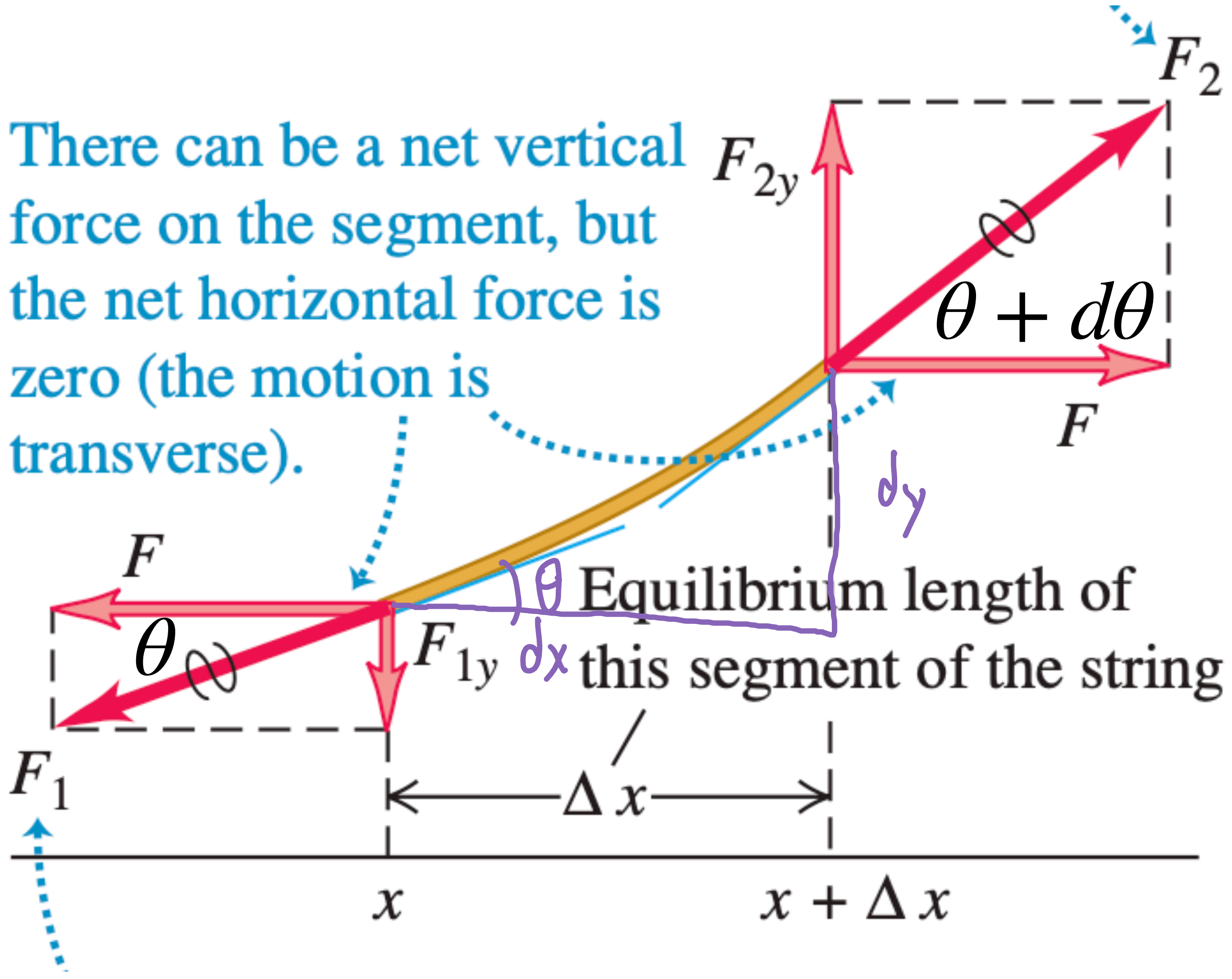
one peak to another



Outline

- Wave equation for transverse wave
- Energy & Power for transverse wave
- Standing wave
- Normal modes of standing wave w/ different boundary conditions
- Energy in standing wave

Wave Equation of Transverse Wave



- We know F_T, μ
- Assumption: amplitude is small, meaning that $\theta \to 0$
- Idea: Newton's 2nd law

$$\begin{cases} \cos\theta = 1 \\ \sin\theta = \tan\theta \end{cases}$$ (small angle)

$$v = \sqrt{\frac{F_T}{\mu}}$$

x: $F_2 \cos(\theta + d\theta) = F_1 \cos\theta$

$F_2 = F_1 = F_T$ (every where are same tension)

y: $F_T \sin(\theta + d\theta) - F_T \sin\theta = \mu \frac{dx}{\cos\theta} \cdot \frac{d^2y}{dt^2}$

$$F_T (\tan(\theta + d\theta) - \tan\theta) = \mu \cdot dx \cdot \frac{d^2y}{dt^2}$$

$$\tan\theta = \frac{dy}{dx}$$

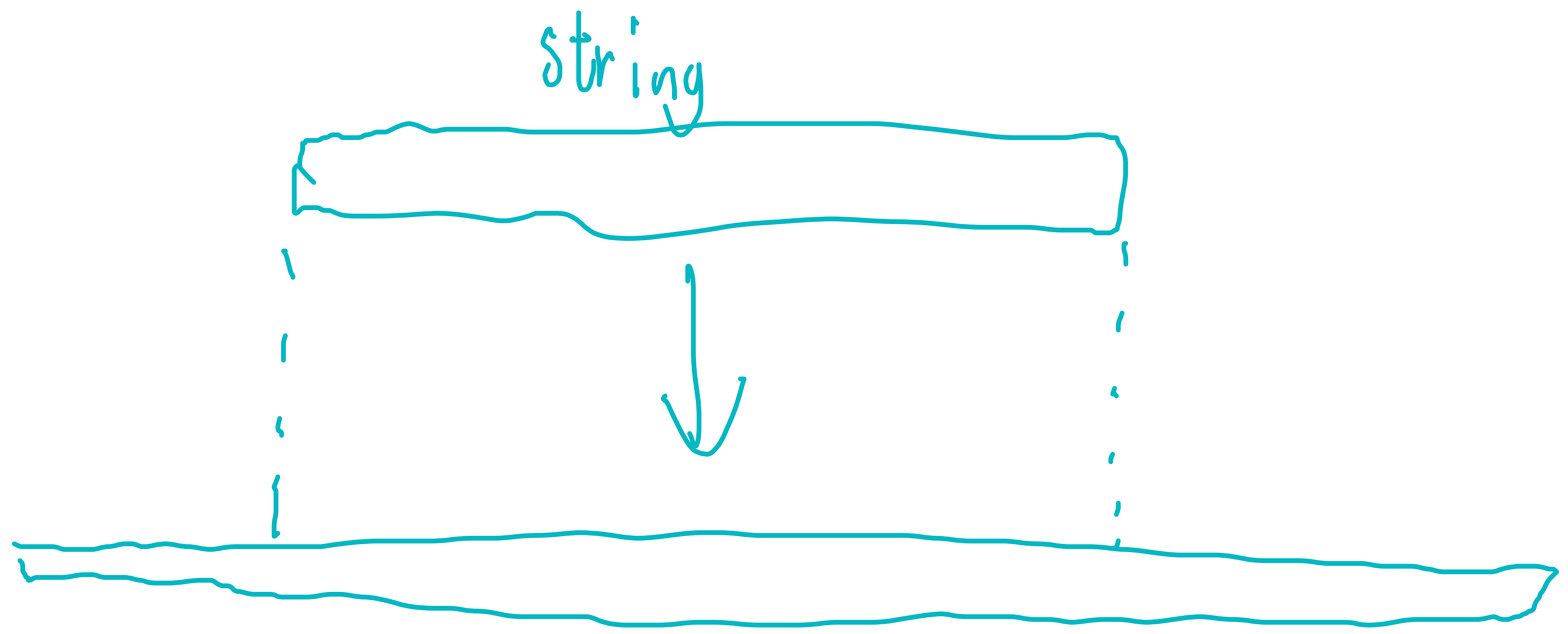
$$F_T \left(\frac{dy}{dx}\Big|_{x+dx} - \frac{dy}{dx}\Big|_{x} \right) = \mu \, dx \, \frac{d^2y}{dx^2}$$

$$F_T \frac{d^2y}{dx^2} = \mu \frac{d^2y}{dt^2}$$

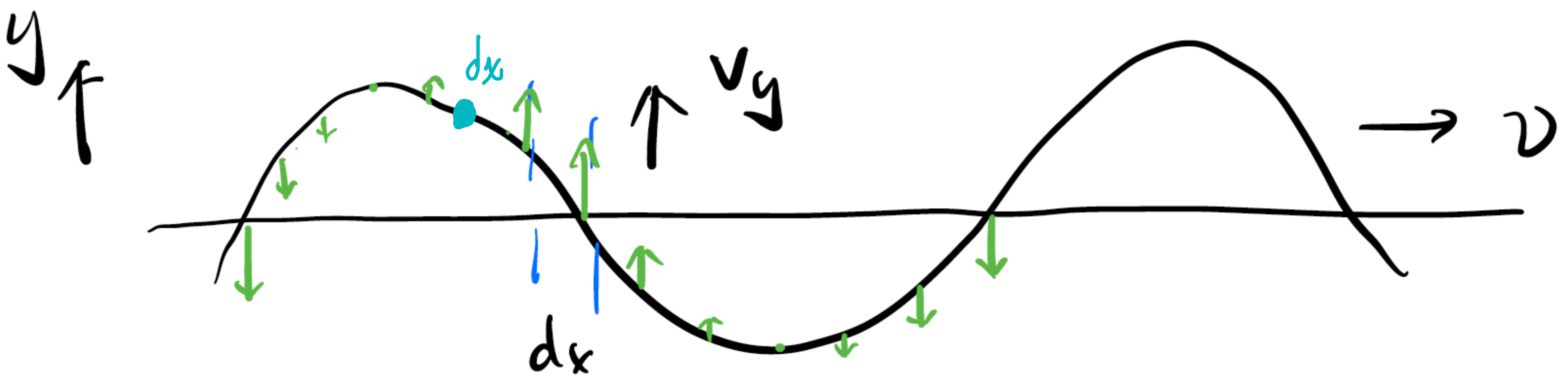
$$\frac{\partial^2 y}{\partial t^2} = \frac{F_T}{\mu} \cdot \frac{\partial^2 y}{\partial x^2}$$

Wave Speed & Wave ~~Function~~ Equation of Transverse Wave

- Wave equaton for a mechanical transverse wave: $\frac{\partial^2 y}{\partial t^2} = \frac{F_T}{\mu}\frac{\partial^2 y}{\partial x^2}$, where
    - μ: mass per unit length, kg/m
    - F_{T}: tension in the string
- Wave speed $v = \sqrt{\frac{F_{\mathrm{T}}}{\mu}}$



Energy & Power in Traveling Wave



- We know
  - $y(x,t) = \cancel{A\sin(kx - \omega t)}$ $A\cos(kx - \omega t)$
  - $v = \sqrt{\frac{F_T}{\mu}}$
- Find E in **one wavelength**.
- Since E=K+U, let's find K in one wavelength..

$$\lambda = vT = \sqrt{\frac{F_T}{\mu}} \cdot \frac{2\pi}{\omega}$$

$$U = \frac{\pi}{2}\sqrt{F_T \mu}\, \omega A^2$$

$$E = k + U$$

(in one wavelength) $= \pi \sqrt{F_T \mu}\, \omega A^2$

$$P = \frac{E}{T} = \frac{\pi\sqrt{F_T\mu}\,\omega A^2}{\frac{2\pi}{\omega}} = \frac{1}{2}\sqrt{F_T \mu}\,\omega^2 A^2$$

$$dk = \frac{1}{2} dm \cdot v_y^2$$

$$v_y = \frac{\Delta y}{\Delta t} = \omega A \sin(kx - \omega t)$$

$$dm = \mu\, dx$$

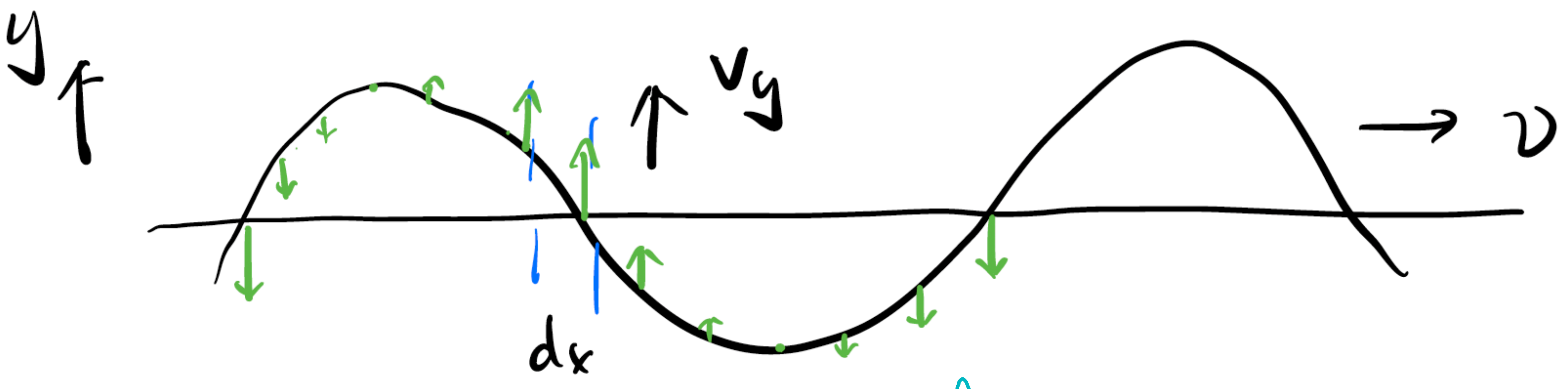
$$k = \int_0^\lambda dk = \frac{1}{2}\mu\, dx\, \omega^2 A^2 \sin^2(kx - \omega t)$$

$$= \frac{1}{2}\mu \omega^2 A^2 \cdot \underbrace{\int_0^\lambda \sin^2(kx - \omega t)\, dx}_{\frac{\lambda}{2}}$$

$$= \frac{1}{4}\mu \omega^2 A^2 \lambda$$

$$= \frac{\pi}{2}\sqrt{F_T \mu}\, \omega A^2$$

Energy & Power in Traveling Wave



$\int \frac{1}{2} k y^2$

string constant

$\omega^2 = \frac{k}{m} = \frac{k_s}{\mu\, dx}$

- We know
  - $y(x,t) = A\sin(kx - \omega t)$
  - $v = \sqrt{\frac{F_T}{\mu}}$
- Find E in **one wavelength**.
- Since E=K+U, let's find U in one wavelength..

Energy & Power in Traveling Wave

- Total energy in one wavelength $E = \pi\sqrt{\mu F_\mathrm{T}}\omega A^2$
  - $K = \dfrac{\pi}{2}\sqrt{\mu F_\mathrm{T}}\omega A^2$
  - $U = \dfrac{\pi}{2}\sqrt{\mu F_\mathrm{T}}\omega A^2$
- Average power $P_\mathrm{av} = \dfrac{E}{T} = \dfrac{1}{2}\sqrt{\mu F_\mathrm{T}}\omega^2 A^2$

$K = U$

Wave Superposition

- Wave superposition (also called interference): $y(x,t) = y_1(x,t) + y_2(x,t)$
- Boundary conditions:
  - Fixed/closed end: $y(x = x_0, t) = 0$, i.e. amplitude A is 0
  - Free/open end: amplitude $A = A_{\max}$

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$$

Standing Wave

- $y(x,t) = 2A\,\cancel{\sin}^{\cos}\,kx\cos\omega t$
- **Node**: amplitude is 0, destructive interference
- **Anti-node**: amplitude is maximum, constructive interference
- Boundary conditions:
  - Fixed/closed end: node

    can't move
  - Free/open end: anti-node

    ↳ move freely

$y_1 = A\cos(kx - \omega t)$, $+x$

$y_2 = A\cos(kx + \omega t)$, $-x$

$\cos(\theta + \alpha) = \cos\theta\cos\alpha - \sin\theta\sin\alpha$

$\xrightarrow{y_1} \quad \xleftarrow{y_2}$

$y = y_1 + y_2 = 2A\underbrace{\cos kx}_{\text{space}}\,\underbrace{\cos\omega t}_{\text{time}}$

$|\cos kx| = 1 \Rightarrow$ max. amplitude

(anti-node)

$\cos kx = 0 \Leftrightarrow kx = (n + \frac{1}{2})\pi$, $k = \frac{2\pi}{\lambda}$

$x = \frac{n + \frac{1}{2}}{2}\lambda$

↳ $y = 0$ (node)

↳ does not osillate

Two Fixed/Closed Ends

- Boundary conditions: $y(x=0,t) = y(x=L,t) = 0$
- Normal modes: $\lambda_n = \dfrac{2L}{n}, n = 1,2,3,...$

L

$x=0$ $y=0$

$x=L$ $y=0$

$A(x) = 2A\cos(kx+\varphi)$

$x=0$

$\cos\varphi = 0 \quad \varphi = \frac{\pi}{2}, \frac{3\pi}{2}$

$\Rightarrow A(x) = 2A\sin kx$

$x=L$

$\sin kL = 0 \quad kL = n\pi$

$\frac{2\pi}{\lambda}\cdot L = n\pi$

$\lambda = \frac{2}{n}\cdot L$

Normal Modes

- $n = 1$, first harmonic/fundamental , 2L
- $n = 2$, second harmonic
- …

Fundamental / 1st harmonic

$n=1$

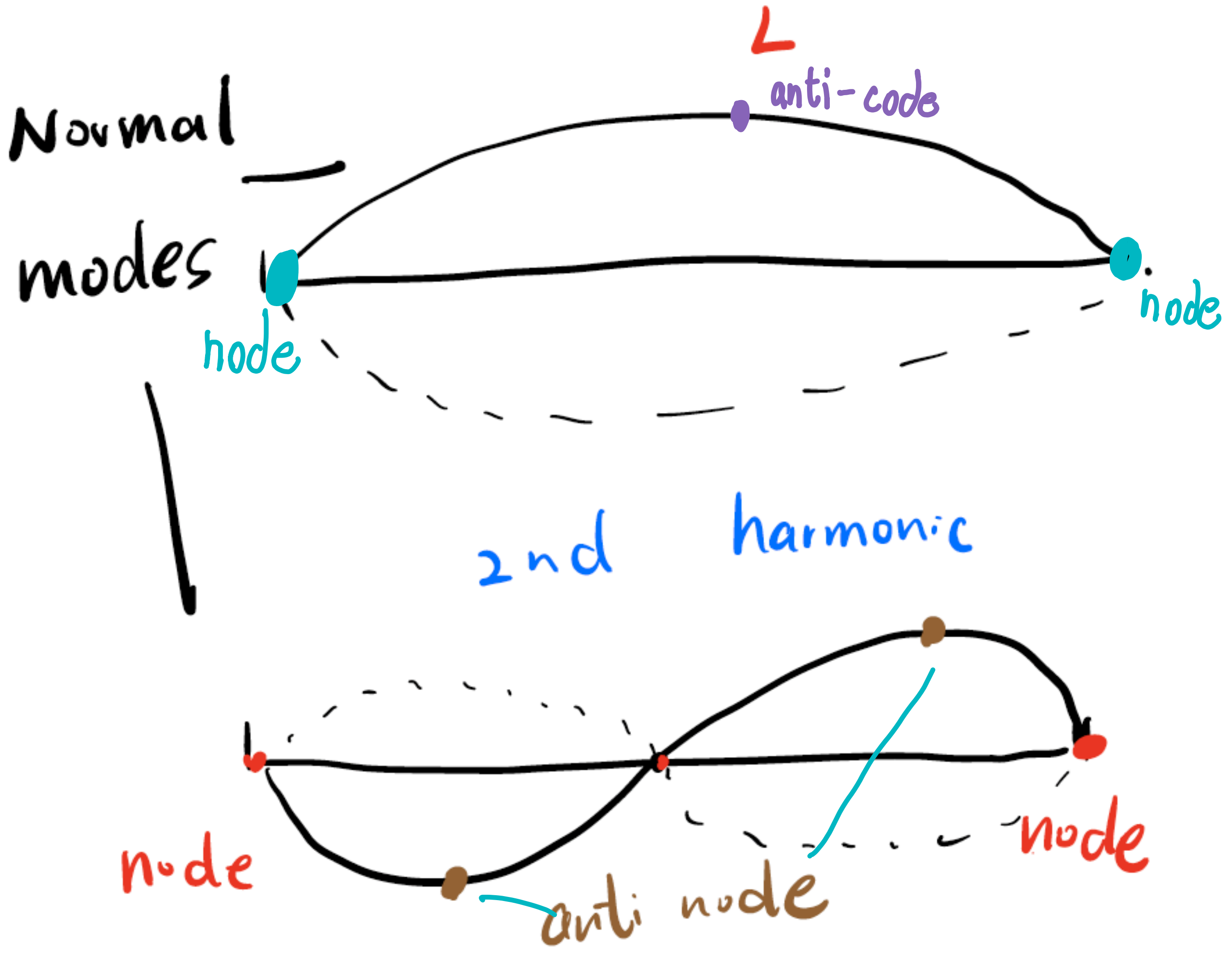
$\lambda = 2L$

$f_1 = \frac{v}{2L}$

$f_n = n f_1$

$\lambda = \frac{2L}{n}$

$n=1$



$n=2$

$\lambda = L$

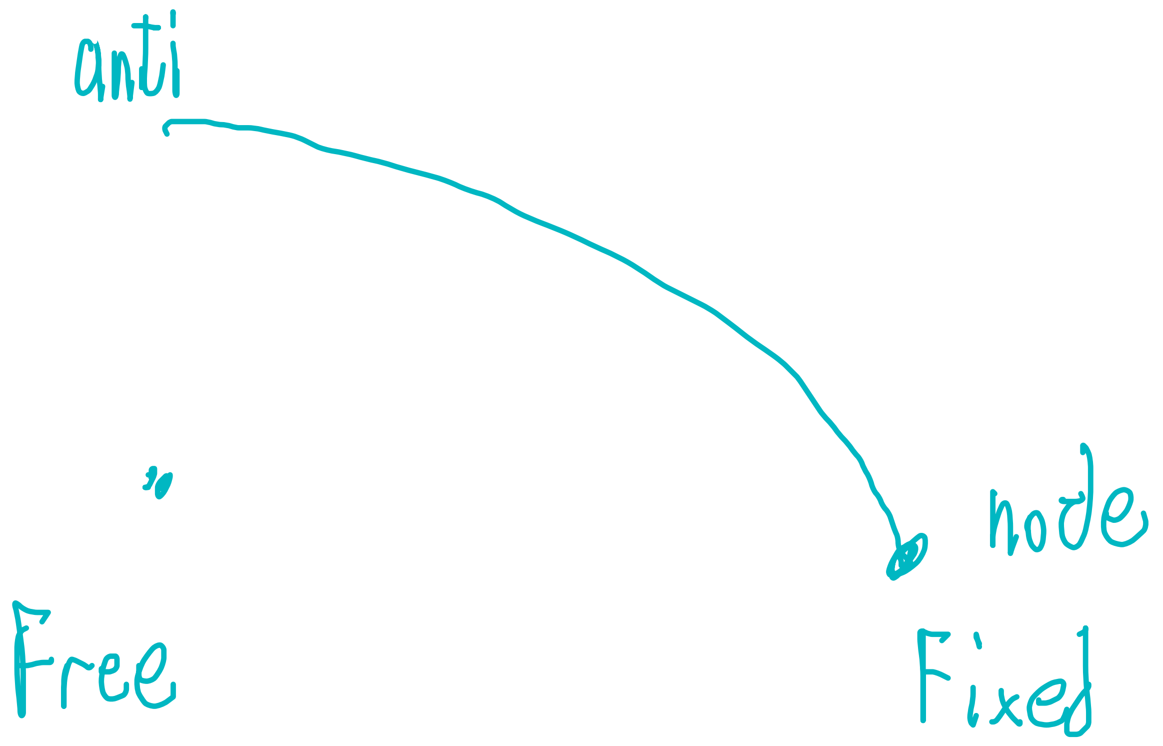
$n=2$

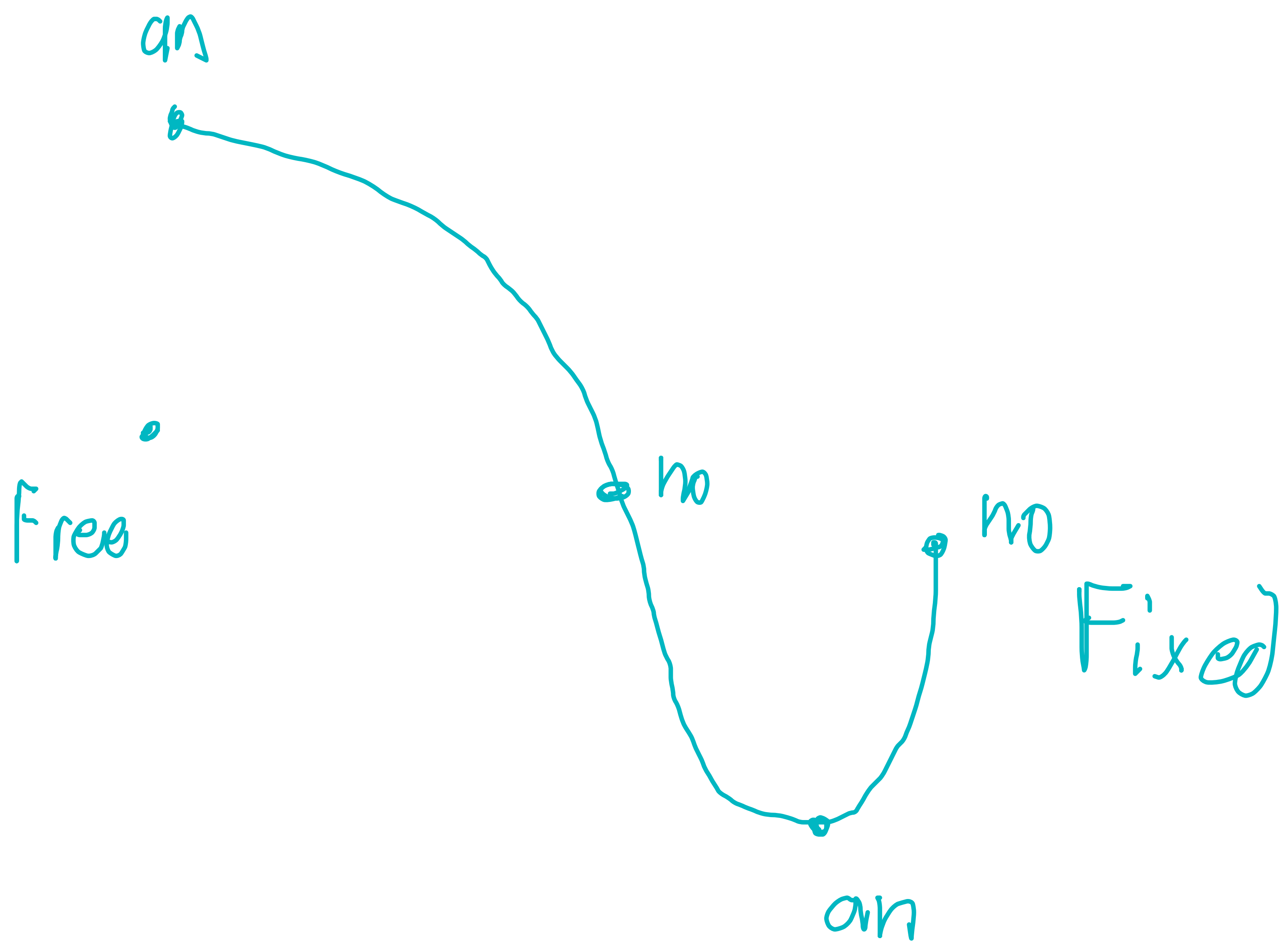
node $\sin kx = 0$

anti-node $\sin kx = \pm 1$

Free & Fixed Ends

- One free end, one fixed end





$n=1$

$n=2$

$\frac{1}{4}\lambda = L$, $\lambda = 4L$

$\frac{3}{4}\lambda = L$ $\lambda = \frac{4}{3}L$

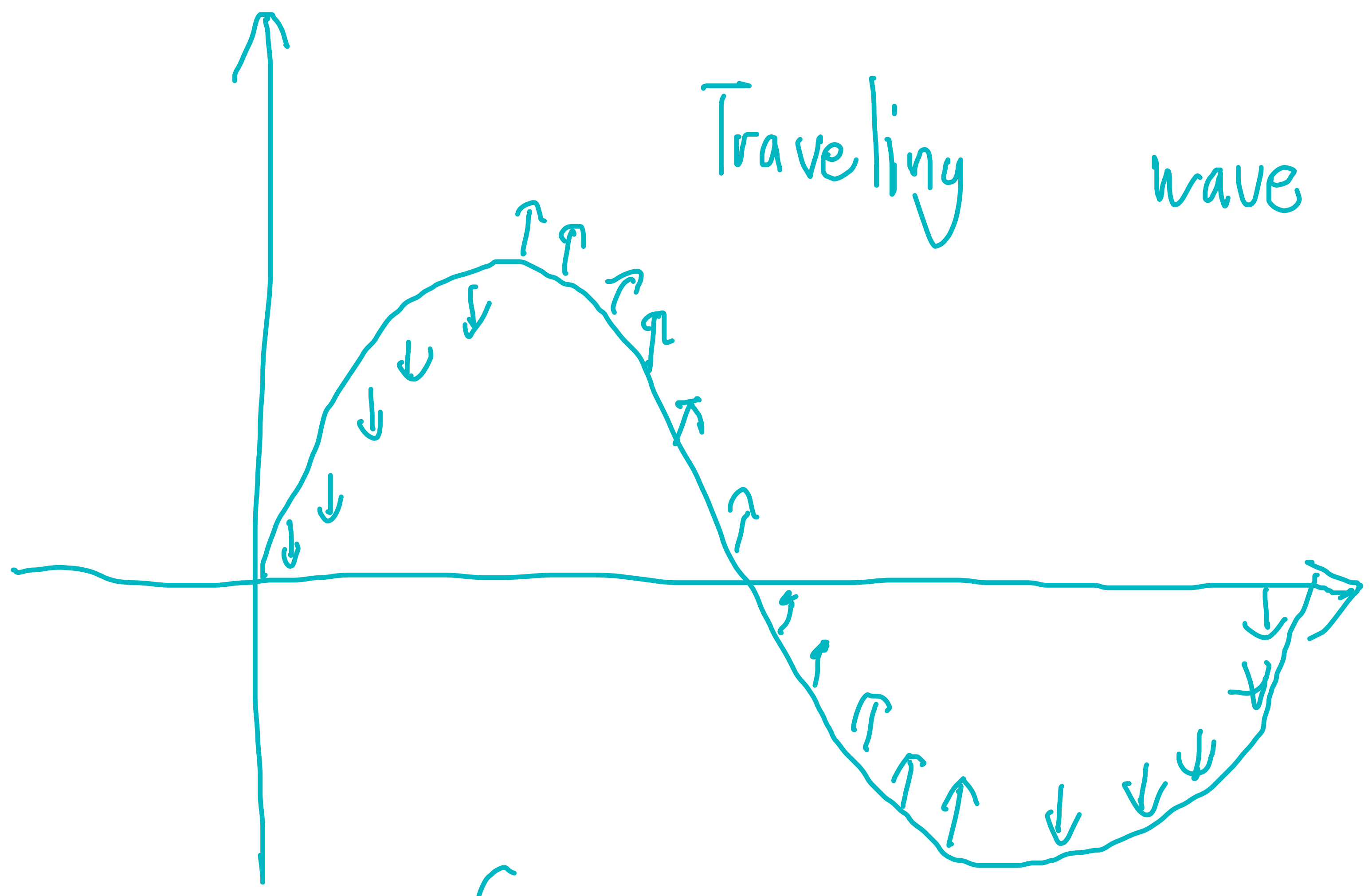
Two Free Ends

- Please work on it yourself.

Energy in Standing Wave

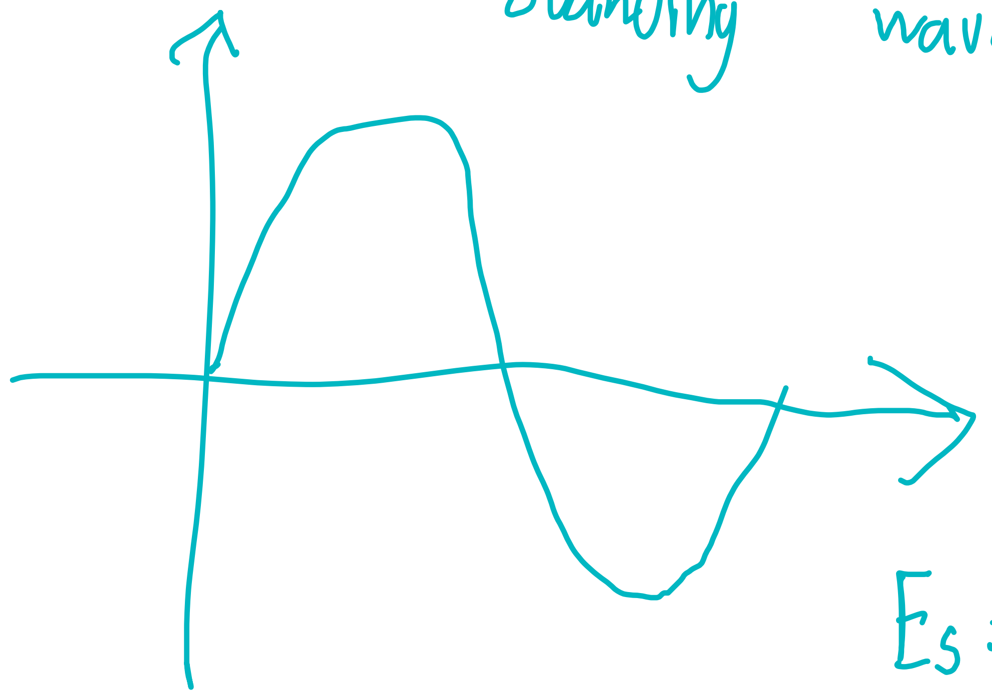
- Energy of standing wave in one wavelength: $E = \frac{\pi}{2}\sqrt{\mu F_{\mathrm{T}}}\omega A^2$
- Is energy conserved when two traveling waves create a standing wave?

Traveling wave



$\begin{cases} E = \pi\sqrt{\mu F_T}\, \omega A^2 \\ K = U = \frac{E}{2} \end{cases}$

standing wave



$V_y = 0$ → just have U, x K

$E_S = U_S = U_T = \frac{E_T}{2} = \frac{\pi}{2}\sqrt{\mu F_T}\,\omega A^2$

$y_1 = A\cos(kx - \omega t)$

$y_2 = A\cos(kx + \omega t)$ $\Rightarrow y = 2A\cos\omega t\cos kx$

$E_{tot} = (A^2 + A^2)\cdot \text{const}$ $\quad E_{tot} = (2A)^2\cdot\frac{1}{2}\cdot\text{const} = 2A^2$